



ĐẠI HỌC
BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY
OF SCIENCE AND TECHNOLOGY

WEEK 1 PROGRESS REPORT

Semi-Supervised Fuzzy C-Means Clustering

Người thực hiện:

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ONE LOVE. ONE FUTURE.

- ▶ Reviewed three papers on sSFCM, sSMC-FCM, and clustering validity.
- ▶ Traced the development from FCM to two forms of semi-supervision.
- ▶ Verified the shared 20-point numerical example and reported memberships.
- ▶ Defined the first implementation and experiment sequence.

Current outcome

Implementation target

sSMC-FCM

with FCM and sSFCM baselines

Week 1 boundary

The work so far covers paper analysis and experiment design. Algorithm code has not yet been implemented.

Standard FCM

$$J_m(U, V) = \sum_{i=1}^N \sum_{k=1}^C u_{ik}^m D_{ik}^2, \quad \sum_{k=1}^C u_{ik} = 1$$

$$V_k = \frac{\sum_{i=1}^N u_{ik}^m x_i}{\sum_{i=1}^N u_{ik}^m}$$

- ▶ u_{ik} : membership of point x_i in cluster k .
- ▶ V_k : centroid of cluster k .
- ▶ $m > 1$: common fuzzification coefficient.
- ▶ $D_{ik} = \|x_i - V_k\|$.

Why add supervision?

A small set of trusted point-to-cluster hints can guide ambiguous samples while the remaining data stay unlabeled.

Core formulation

$$J(U, V) = \sum_{i=1}^N \sum_{k=1}^C |u_{ik} - \bar{u}_{ik}|^m D_{ik}^2$$

$$u_{ik} = \bar{u}_{ik} + \left(1 - \sum_{j=1}^C \bar{u}_{ij}\right) \frac{D_{ik}^{-2/(m-1)}}{\sum_{\ell=1}^C D_{i\ell}^{-2/(m-1)}}$$

- ▶ $\bar{U}$ stores membership grades supplied in advance.
- ▶ Missing supervision is represented by $\bar{u}_{ik} = 0$.
- ▶ $\bar{U}$ changes the membership update directly.
- ▶ Centroids use weights $|u_{ik} - \bar{u}_{ik}|^m$.
- ▶ The algorithm alternates membership and centroid updates.

Objective

$$J(U, V) = \sum_{i=1}^N \sum_{k=1}^C u_{ik}^{m_{ik}} D_{ik}^2$$

- ▶ No auxiliary supervised membership matrix.
- ▶ Unsupervised memberships use the standard FCM update.
- ▶ A supervised membership solves Eq. (19), then normalizes the result.

How supervision changes m_{ik}

Unsupervised point x_i
 $m_{ij} = M$ for every cluster j

↓ add target-cluster information

Point x_i supervised toward cluster k
 $m_{ik} = M_0 > M$, and
 $m_{ij} = M$ for $j \neq k$

Effect: The selected exponent changes both membership and centroid weighting. Equation (23) guides M_0 for a target $u_{ik} \geq \alpha$. Here M_0 denotes the paper's M' .

Aspect	sSFCM 2009	sSMC-FCM 2021
Supervision	Values in $\bar{U}$	Target pairs in a supervised set
Objective	$ u_{ik} - \bar{u}_{ik} ^m D_{ik}^2$	$u_{ik}^{m_{ik}} D_{ik}^2$
Main parameter	Supervised grade $\bar{u}_{ik}$	Coefficients M and $M_0 > M$
Membership update	Adds supervised grade, then allocates the remaining mass	Standard FCM for unlabeled points; scalar solve for a target pair
Centroid influence	Weight $ u_{ik} - \bar{u}_{ik} ^m$	Weight $u_{ik}^{m_{ik}}$
Extra input	Auxiliary matrix $\bar{U}$	Supervised point-cluster pairs and M_0
Main idea	Modify membership information	Modify fuzzification behavior

Implementation consequence

The 2021 method replaces direct membership guidance with pair-specific exponents and a numerical membership subproblem.

Shared 20-Point Numerical Example

Method	Supervision	U_{C1}	U_{C2}	Hard cluster
FCM	none	0.19	0.81	C2
sSFCM 2009	$\bar{u}_{i1} = 0.3$	0.45	0.55	C2
sSFCM 2009	$\bar{u}_{i1} = 0.6$	0.69	0.31	C1
sSMC-FCM 2021	$M = 2, M_0 = 4$	0.43	0.57	C2
sSMC-FCM 2021	$M = 2, M_0 = 8$	0.60	0.40	C1

Points 9 and 10 have the same reported memberships

Both mechanisms progressively pull the supervised samples toward cluster 1. Stronger supervision changes their hard assignment from cluster 2 to cluster 1.

Interpretation limit

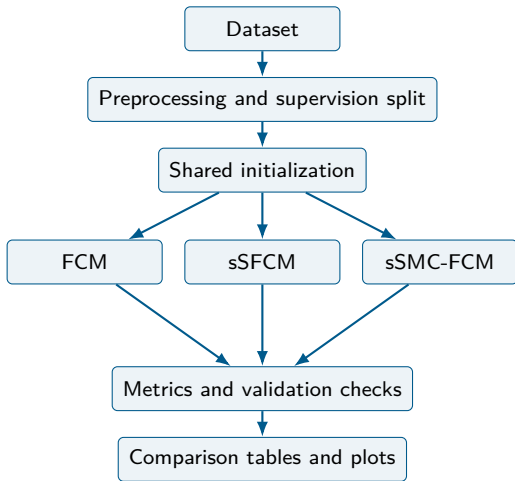
This example demonstrates comparable behavior. It does not establish general superiority of the 2021 method.

Scoring rule: The 2010 criteria evaluate hard partitions, so use $\hat{y}_i = \arg \max_k u_{ik}$ before scoring.

Ground truth	Metric	What it measures	Better
Available	ARI	Pairwise agreement, corrected for chance	Higher
Available	Jaccard	Agreement on point pairs assigned together	Higher
Unavailable	SWC / SSWC	Cohesion versus separation; SSWC uses centroids	Higher
Unavailable	VRC	Between-cluster dispersion relative to within-cluster dispersion	Higher

Selection: ARI and Jaccard support labeled benchmarks. Silhouette was among the strongest criteria in the paper, while VRC provides a fast complementary check.

Implementation and Experiment Pipeline



Planned modules

`src/fcm.py`
`src/ssfcm.py`
`src/ssmc_fcm.py`
`src/metrics.py`
`src/datasets.py`
`src/experiments.py`

Implementation order

1. FCM baseline and tests
2. 2009 toy data
3. sSFCM reproduction
4. sSMC-FCM Eqs. (17)–(20)
5. 2021 Table 3 reproduction
6. Metrics and larger data sets

Next coding steps

1. Implement FCM with shape, sum-to-one, and finite-value tests.
2. Recreate the 20-point data set and FCM memberships.
3. Implement sSFCM and reproduce the 2009 values.
4. Implement the supervised sSMC-FCM membership solver.
5. Reproduce the 2021 values before larger experiments.

Questions for discussion

- ▶ Solver tolerance, step size, and upper bound for Eq. (19).
- ▶ Center initialization and stopping norm.
- ▶ Zero-distance and empty-cluster handling.
- ▶ Real benchmark data sets and supervision ratios.

Week 2 gate: reproduce the point 9/10 memberships before testing new data.



Y. Endo, Y. Hamasuna, M. Yamashiro, and S. Miyamoto, “On Semi-Supervised Fuzzy c-Means Clustering,” *IEEE International Conference on Fuzzy Systems*, 2009.



T. D. Khang, M.-K. Tran, and M. Fowler, “A Novel Semi-Supervised Fuzzy C-Means Clustering Algorithm Using Multiple Fuzzification Coefficients,” *Algorithms*, vol. 14, no. 9, article 258, 2021. DOI: 10.3390/a14090258.



L. Vendramin, R. J. G. B. Campello, and E. R. Hruschka, “Relative Clustering Validity Criteria: A Comparative Overview,” *Statistical Analysis and Data Mining*, vol. 3, pp. 209–235, 2010. DOI: 10.1002/sam.10080.

A decorative graphic on the left side of the slide. It features a dark blue background with a large, stylized circular pattern composed of many small red dots. The dots are arranged in concentric, slightly offset rings, creating a sense of depth and movement. The word "HUST" is centered within this pattern.

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THANK YOU !



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